Inverted MAML Card 1: 13-Min 45% PWM 2-Beat Analog Bit-Slicing Thermal Report

Hardware: Card 1 (Trion T120 + T20, 1k/3.3k Attenuator) | **Params:** 45% HW PWM, K=1, LR=0.06, L2=1024, SGD (Mom=0) | **Pairs:** 5,278

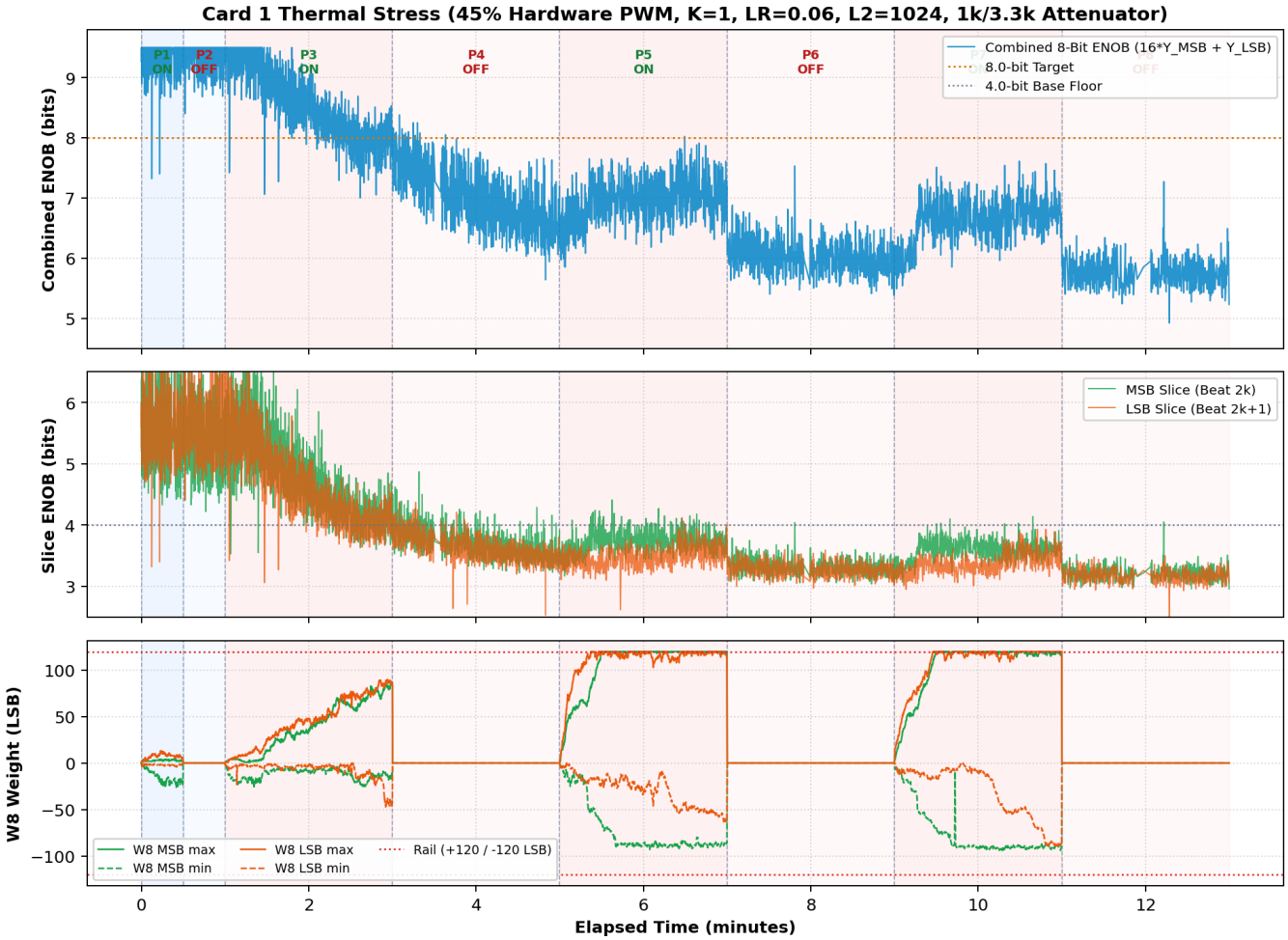
1. Executive Summary

Under sustained 45% hardware PWM thermal stress, open-loop analog precision (Regler OFF, W8=0) dropped by **3.63 bits** from the cold baseline (**9.38 bits** in Phase 2) to **5.75 bits** in Phase 8. Enabling closed-loop Forward Hadamard Propagation (Regler ON, LR=0.06, L2=1024, K=1) delivered clear, repeatable positive ENOB steps:

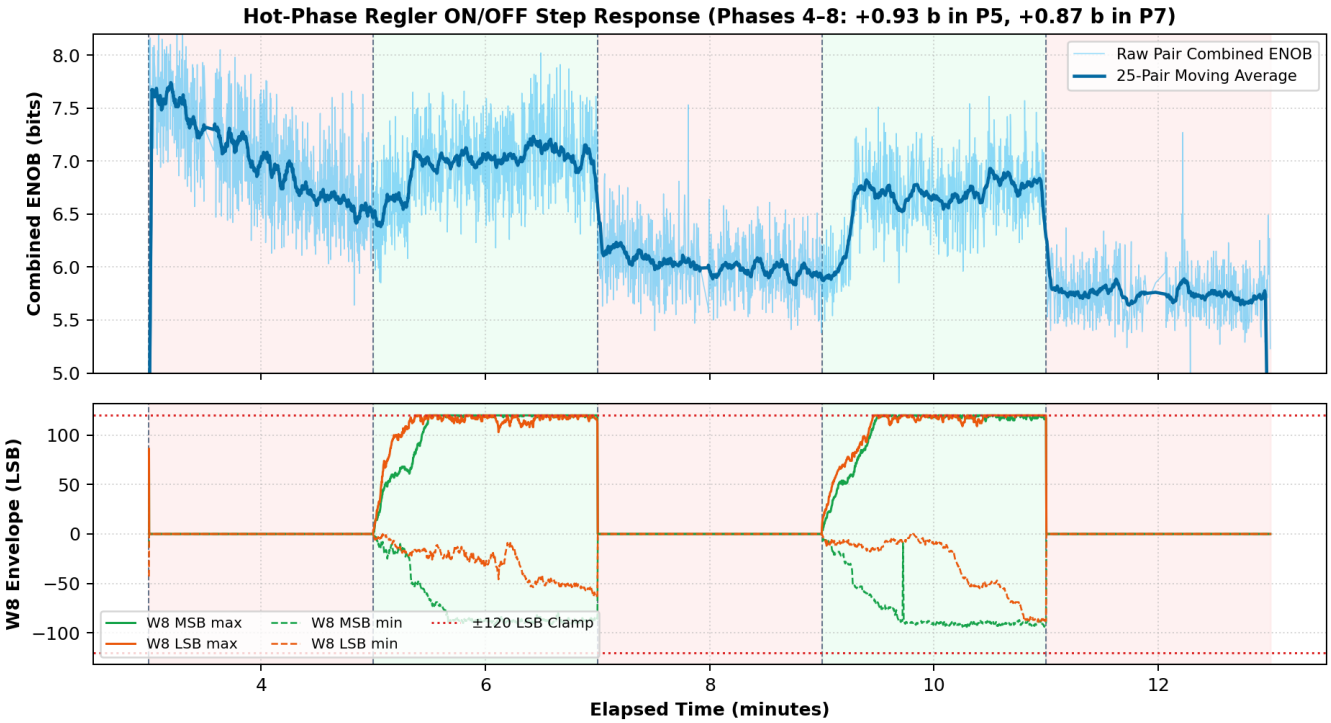
- **Phase 5 vs Phase 6 (Warm):** Mean **+0.93 bits** (6.95 b vs 6.02 b); steady-state **+1.08 bits** (7.06 b vs 5.98 b).
- **Phase 7 vs Phase 8 (Warm):** Mean **+0.87 bits** (6.62 b vs 5.75 b); steady-state **+1.01 bits** (6.74 b vs 5.73 b).
- **Physical Bottleneck:** Both W8_msb and W8_lsb **railed at +120 LSB** in Phases 5 and 7 under the 1k/3.3k attenuator, capping compensation authority. Resoldering to **1k/2.2k** (+50% Vernier gain) is the next hardware step.

2. Phase-by-Phase Performance (5,278 Pairs)

Phase	Regime	N	Mean Comb	SS Comb	MSB	LSB	W8 MSB [min,max]	W8 LSB [min,max]
P1	Cold / Regler ON	481	9.37 b	9.36 b	5.51 b	5.58 b	[-26, +5]	[-5, +13]
P2	Cold / Regler OFF	387	9.38 b	9.36 b	5.38 b	5.62 b	[+0, +0]	[+0, +0]
P3	Warm / Regler ON	1208	8.59 b	8.14 b	4.78 b	4.65 b	[-26, +88]	[-48, +90]
P4	Warm / Regler OFF	858	7.03 b	6.73 b	3.71 b	3.63 b	[+0, +0]	[+0, +0]
P5	Warm / Regler ON	690	6.95 b	7.06 b	3.72 b	3.48 b	[-93, +120]	[-63, +120]
P6	Warm / Regler OFF	616	6.02 b	5.98 b	3.31 b	3.27 b	[+0, +0]	[+0, +0]
P7	Warm / Regler ON	549	6.62 b	6.74 b	3.59 b	3.37 b	[-94, +120]	[-89, +120]
P8	Warm / Regler OFF	489	5.75 b	5.73 b	3.21 b	3.17 b	[+0, +0]	[+0, +0]


Card 1 Thermal Stress (45% Hardware PWM, K=1, LR=0.06, L2=1024, 1k/3.3k Attenuator)
P1 ON
P2 OFF
P3 ON
P4 OFF
P5 ON
P6 OFF
Combined 8-Bit ENOB (16*Y_MSB + Y_LSB)
8.0-bit Target
4.0-bit Base Floor
Combined ENOB (bits)
MSB Slice (Beat 2k)
LSB Slice (Beat 2k+1)
Slice ENOB (bits)
W8 MSB max
W8 MSB min
W8 LSB max
W8 LSB min
Rail (+120 / -120 LSB)
W8 Weight (LSB)
Elapsed Time (minutes)

3. Hot-Phase Step Response & Weight Saturation Diagnosis



In Phases 5 and 7, activating closed-loop SGD immediately lifts Combined ENOB by ~0.9–1.1 bits relative to the adjacent Regler-OFF ablation phases (P6 and P8). However, the weight envelope plot confirms that the positive W8 rail (+120 LSB) is reached on both MSB and LSB slices. Because the 1k/3.3k attenuator limits the Vernier branch current injection into SUM35, further thermal drift cannot be absorbed once W8 saturates.

4. Bit-Slicing Architecture & Post-Analog Alpha/Beta Roadmap

- **2-Beat Time-Multiplexed Slicing:** Beat 2k (MSB, Bank 0) and Beat 2k+1 (LSB, Bank 1, 0.75x scale) run with fully isolated T120 weight/velocity banks (0x1E subcmd 4) and locked t=0 cold alpha/beta targets (0x1E subcmd 5).
- **Immediate Hardware Action (1k/2.2k Attenuator):** Replacing the 3.3k summing resistor with 2.2k increases Vernier authority by 1.50x, giving W8 headroom to compensate 45% thermal drift without railing at +120 LSB.
- **8-Channel FHP Post-Analog Stage (PCBA v1.3.2 / SKY130):** Per *post_analog_alpha_beta_calibration_8_channel_fhp_specification.md*, expanding the Hadamard burst from 12 to 16 pulses (Ch 0–5 matrix rows, Ch 6 V_gain shunt-triode, Ch 7 V_offset) offloads static DC offset and linear gain from Matrix 3/8, allowing L2 regularization to keep W8 centered near 0 LSB for full non-linear thermal headroom.